

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained real-time networking infrastructure and device <-> cloud communication systems for access control hardware, ensuring 99.95% uptime.
- Implemented robust device identity and authentication mechanisms using short-lived tokens and mutual TLS, significantly reducing unauthorized access incidents by **40%**.
- Managed large-scale Linux device fleet updates and rollouts using automated orchestration pipelines, successfully deploying critical patches to over **5,000 devices** without service interruption.
- Designed and implemented event-driven backend services using Python and Flask to handle asynchronous device status updates and command processing, improving system responsiveness by **30%**.
- Enhanced system observability by integrating **Splunk** and **CloudWatch** for log aggregation and real-time performance monitoring of distributed controller systems, reducing mean time to detect (MTTD) by **25%**.
- Automated critical production support tasks using Python scripts, reducing manual intervention for incident response and system diagnostics by **50%**.

SKILLS

Backend & Systems – Distributed Systems, Event-Driven Architecture, Real-Time Networking, Device Fleet Management, Authentication, Security, API Design, System Reliability, Observability, Debugging, Production Tooling, Device Update Orchestration

Languages & Scripting – Python, Java, SQL, Bash, TypeScript

Frameworks & Libraries – Spring Boot, Flask, FastAPI, Node.js, NestJS, pandas, NumPy, scikit-learn

Databases – PostgreSQL, MySQL, Oracle

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Linux Systems

CI/CD & DevOps – Jenkins, Git, Code Reviews

Data & Analytics – Power BI, Tableau, DAX, Power Query, Data Modelling, ETL

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Distributed Access Control Platform

- Engineered a scalable backend API using **Java Spring Boot** and **PostgreSQL** to manage device registration, authentication, and real-time status updates for a fleet of embedded Linux controllers.
- Implemented distributed coordination logic to ensure reliable command execution and state synchronization across devices operating under unreliable network conditions, achieving 99.9% command delivery success

rate.

- Developed automated CI/CD pipelines with **Jenkins** and **Docker** for consistent deployment of backend services and device firmware updates.

Real-Time IoT Telemetry Dashboard

- Built an interactive dashboard using **Power BI** and **DAX** to visualize real-time telemetry data from connected devices, enabling operations teams to monitor system health and identify performance bottlenecks.
- Developed Python scripts for data ingestion and transformation, processing over 1 million data points daily from diverse device sources into a centralized data warehouse.
- Integrated with **AWS** services including S3 for data storage and Lambda for event-driven data processing, ensuring a cost-effective and scalable data pipeline.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)